

Gaussian Diffusion vs. Normalizing Flows

(concise overview)

Aspect	Diffusion / Score-based	Normalizing Flows
Core training target	Denoiser / score over noise levels ¹	Change-of-variables likelihood ²
Likelihood	Typically intractable (bounds/ODE exceptions) ³	Exact log-likelihood & bpd ⁴
Sampling cost	Many steps; ~ 35 feasible with EDM ⁵ ; faster in latent space ⁶	Single pass (fast)
Invertibility	No	Yes (exact inverse)
Conditioning	Very flexible (CFG, cross-attention) ⁶	Via conditioning in coupling/1×1 conv; integrates into likelihood
Strengths	SOTA perceptual quality; robust training	Exact density; calibrated comparisons; natural for cross-modal transforms
Typical pitfalls	Slow sampling; compute-heavy; likelihood metrics awkward	Architecture care needed; coupling expressivity vs. Jacobian cost

- **Modeling objective**

- Diffusion / score-based: learn time-indexed score/denoiser for a noise-perturbed data process; sampling integrates a reverse (S)DE or a discrete Markov chain over many steps¹.
- Flows: learn a bijective map with the change-of-variables formula, giving exact log-likelihoods and exact latent inference in one pass².

- **Likelihood & calibration**

- Diffusion: likelihood typically intractable (exceptions via variational bounds or specialized ODE setups); models are often tuned for perceptual quality¹.

- Flows: tractable log-likelihood and bits-per-dim out-of-the-box; useful for model selection and uncertainty that relates to density⁷.
- **Sampling cost**
 - Diffusion: tens to thousands of network evaluations per sample; modern designs bring this to ~35 evals while preserving quality⁵. Latent Diffusion further cuts cost by operating in autoencoder latent space⁶.
 - Flows: 1 forward pass from base noise to data (fast synthesis)⁴.
- **Invertibility and cross-modal mapping**
 - Diffusion: not inherently invertible; inverse problems solved via guidance or conditional score estimation⁸.
 - Flows: exactly invertible by design, enabling clean cross-modal synthesis and level-wise latent taps (helpful for our per-level alignment and imputation).
- **Conditioning & control**
 - Diffusion: rich conditioning (classifier-free guidance, text/image/ROI prompts) and strong high-fidelity synthesis⁶.
 - Flows: conditioning via invertible affine/1x1 conv blocks or coupling networks; conditioning enters the likelihood cleanly⁴.
- **Memory & training dynamics**
 - Diffusion: training is stable; losses are local denoising/score terms; large compute budgets often used¹.
 - Flows: training can be trickier (Jacobian/ActNorm/coupling design), but architectures like Glow offer reversible layers and stable multi-scale training when implemented carefully⁴.
- **Medical-imaging relevance**
 - Diffusion: excellent for inpainting/denoising/segmentation priors via guidance; strong perceptual realism.
 - Flows: exact likelihoods + invertibility make them natural for multimodal latent alignment and our Conditional Gaussian imputation pipeline.

References

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